

1. Cloud Computing

What is it?

Cloud computing means using computing resources—such as servers, databases, storage and software—over the internet instead of relying entirely on computers and servers located within the organisation.

Simple example

Instead of a school purchasing and maintaining its own physical server to store student records, it could use a cloud platform to store the data and access it through the internet.

Real-world examples

Netflix

- Netflix stores and delivers its huge library of movies and TV shows using cloud infrastructure.
- When millions of users watch videos simultaneously, cloud resources can scale to handle demand.

Google Drive / Microsoft 365

- Documents are stored in the cloud.
- Users can access them from laptops, tablets or mobile phones.
- Multiple people can collaborate on the same document.

Business example

Imagine an **e-commerce company**.

Traditional approach:

Buy server → install server → maintain server → upgrade hardware → increase capacity

Cloud approach:

Subscribe to cloud infrastructure → increase resources when traffic increases → reduce resources when traffic falls

For example, during **11.11 or Black Friday**, an online store may experience a huge increase in website visitors. Cloud computing allows the company to scale its infrastructure to cope with the increased traffic.

Main benefits

Benefit	Explanation
Scalability	Resources can increase or decrease according to demand
Lower infrastructure costs	Less need to purchase physical servers
Accessibility	Systems can be accessed from different locations
Collaboration	Employees can work on shared cloud applications
Backup	Data can be backed up to cloud infrastructure

Easy way to remember:

Cloud = computing resources available through the internet.

2. Artificial Intelligence (AI) 🤖

What is it?

Artificial Intelligence enables computers to perform tasks that normally require human intelligence, such as:

- analysing information
- recognising patterns
- making predictions
- understanding language
- generating content
- recommending products
- automating decisions or workflows

Real-world example: Netflix recommendations

When Netflix recommends:

"Because you watched..."

AI analyses information such as:

- what you watched
- how long you watched
- what you searched for

- what similar users watched

It then predicts what you might be interested in.

Business example: E-commerce

Imagine an online fashion store.

A customer searches for:

"Men's running shoes"

AI can analyse the customer's behaviour and recommend:

- running socks
- sports shirts
- fitness watches
- other running shoes

The company therefore uses AI to **personalise the customer's shopping experience**.

Another example: Banking

Banks can use AI to detect unusual transactions.

For example:

Customer normally spends \$50–\$200 in Singapore.

Suddenly:

\$4,500 transaction occurs overseas.

An AI system can identify this as an unusual pattern and trigger an alert for possible fraud.

AI in education

A school could use AI to:

- analyse student performance
- identify students who may need additional support
- generate practice questions
- provide personalised learning recommendations
- automate administrative tasks
- analyse feedback from students

Main benefits

Benefit	Example
Automation	Automatically process repetitive tasks
Prediction	Forecast customer demand
Personalisation	Recommend products/content
Pattern recognition	Detect fraud or unusual behaviour
Decision support	Provide insights to managers

Easy way to remember:

AI = technology that analyses information and performs tasks that normally require human intelligence.

3. Internet of Things (IoT)

What is it?

The **Internet of Things** refers to physical objects that contain:

- sensors
- software
- connectivity

These objects can collect and transmit data through a network.

In simple terms:

Physical object → Sensor → Internet/network → Data → Action

Real-world example: Smart agriculture

The image itself shows **smart agriculture**.

Imagine a farmer has sensors installed in a field.

The sensors measure:

- soil moisture
- temperature
- humidity

- sunlight
- water levels

Suppose the soil becomes too dry.

The sensor sends:

Soil moisture = 20%

The system can automatically activate irrigation.

So instead of the farmer manually checking every field:

Sensor → Data → System → Automatic irrigation

Real-world example: Smart warehouse

A logistics company can place IoT sensors on goods and equipment.

For example:

Temperature sensor → Warehouse system

If the temperature in a refrigerated storage area becomes too high:

Sensor detects 12°C → System sends alert → Staff investigates

This is particularly useful for:

- food
- medicine
- vaccines
- fresh produce

Real-world example: Smart manufacturing

A factory can install sensors on machines.

The sensors continuously measure:

- temperature
- vibration
- operating speed
- energy consumption

Suppose a machine normally operates with a vibration level of 10.

The sensor detects:

Vibration = 25

The system can alert maintenance staff before the machine completely fails.

This is called **predictive maintenance**.

Main benefits

Benefit	Explanation
Real-time monitoring	Information is collected continuously
Automation	Devices can trigger actions automatically
Predictive maintenance	Problems can be detected before failure
Better efficiency	Resources can be used more effectively
Visibility	Managers can see what is happening remotely

Easy way to remember:

IoT = physical objects collecting and exchanging data through networks.

4. Blockchain

What is it?

Blockchain is a technology for maintaining a shared digital record of transactions or information.

A blockchain records information in a sequence of blocks. Once information is recorded and confirmed, changing the historical record is designed to be difficult.

A key idea is:

Shared record + verification + traceability

Real-world example: Supply Chain

Imagine a company sells organic coffee.

The coffee passes through:

Farmer → Processor → Distributor → Warehouse → Retailer → Customer

Each stage can record information about the product.

For example:

Stage	Information
Farmer	Where coffee was produced
Processor	Processing date
Distributor	Shipment information
Warehouse	Storage information
Retailer	Product received
Customer	Product purchased

Blockchain can provide a shared record that helps different participants verify the product's history.

Why is this useful?

Suppose a company claims:

"This product is ethically sourced."

A traceable digital record can help provide evidence about where the product came from and how it moved through the supply chain.

Real-world example: Digital transactions

Blockchain is also the underlying technology associated with cryptocurrencies such as Bitcoin.

Transactions are recorded on a distributed ledger and verified through the network's consensus mechanisms.

Smart contracts

Blockchain can also support **smart contracts**.

A smart contract is essentially a programmed set of rules.

For example:

IF delivery is confirmed
THEN release payment

Instead of employees manually checking every condition, the system can execute the programmed rule when the required conditions are satisfied.

Main benefits

Benefit	Explanation
Traceability	Track the history of transactions/products
Transparency	Participants can verify recorded information
Tamper resistance	Historical records are difficult to alter
Decentralisation	Records can be shared across participants
Automation	Smart contracts can execute predefined rules

Easy way to remember:

Blockchain = a shared, verifiable and traceable digital record.

5. How These Technologies Work Together

The most important concept for students is that **these technologies can be combined**.

Consider a **smart logistics company**.

Step 1 — IoT collects information

Sensors are attached to delivery trucks.

They collect:

- GPS location
- temperature
- fuel consumption
- vehicle condition



Step 2 — Cloud stores the information

The sensor data is transmitted to a cloud platform.

IoT → Internet → Cloud

The company can access the information from a dashboard.



Step 3 — AI analyses the information

AI analyses the data and discovers:

"Truck #25 is consuming significantly more fuel than normal."

Or:

"The vehicle is likely to require maintenance soon."



Step 4 — Blockchain records important transactions

Blockchain can be used to record important supply-chain events:

Product dispatched → Product received → Delivery confirmed

This provides a traceable record shared among authorised participants.

The overall picture

IoT

Collects data



Cloud Computing

Stores and processes data



AI

Analyses data and generates insights



Blockchain

Provides trusted and traceable records for selected transactions



Business

Makes faster and more informed decisions

6. One Complete Real-World Scenario: Smart Supermarket

Imagine a modern supermarket.

IoT

Sensors monitor:

- refrigerator temperature
- inventory levels
- customer traffic
- energy consumption

Cloud Computing

All the information is sent to a cloud platform.

Managers can view the supermarket's operations from a dashboard.

AI

AI analyses the data and predicts:

"Milk sales will increase by approximately 30% this weekend."

The supermarket can therefore order additional stock.

AI can also identify products that are selling slowly.

Blockchain

For selected products, blockchain could record supply-chain information:

Farm → Distributor → Warehouse → Supermarket

Customers could potentially scan a product and view verified supply-chain information.

Result

Instead of simply **reacting to problems**, the supermarket becomes more **data-driven and proactive**.

7. Quick Comparison for Students

Technology	Main Purpose	Simple Example
 Cloud Computing	Store and provide computing resources	Google Drive
 AI	Analyse, predict and automate	Product recommendations
 IoT	Collect real-world data	Smart temperature sensor
 Blockchain	Record and verify transactions	Supply-chain tracking

Simple memory trick

IoT = Collect

Cloud = Store & Process

AI = Analyse & Predict

Blockchain = Verify & Trace

These four technologies together form an important foundation for **digital transformation**, because they allow organisations to move from manual, reactive operations toward **connected, automated, data-driven operations**.